LOCAL FOOD WASTAGE MANAGEMENT SYSTEM

Business Analytics Project Report

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Technologies Used:

• MySQL

• Python

• Streamlit

• GitHub

Project Scope

• SQL Analysis and Business Insights

• Exploratory Data Analysis (EDA)

• Interactive Streamlit Dashboard

• GitHub Repository Management

• Cloud Deployment using Streamlit

Project Deadline: June 22,2026

Project Deliverables:

✓ SQL Queries & Analysis

✓ Python EDA Notebook

✓ Interactive Dashboard

✓ GitHub Repository

✓ Live Deployed Application

Project Overview

Food wastage is a major challenge that affects both economic resources and food availability. The Local Food Wastage Management System was developed to analyze food donation, distribution, and claim data in order to identify patterns, improve food allocation efficiency, and reduce food wastage.

The project integrates data analytics techniques using SQL, Python, and Streamlit to generate meaningful business insights and visualize key performance indicators through an interactive dashboard.

Project Objectives

- Analyze food donation and distribution data.
- Identify the most active food providers and receivers.
- Monitor food listings and claim activities.
- Track claim completion and distribution performance.
- Visualize trends and patterns through interactive dashboards.
- Support data-driven decision making to reduce food wastage.

Dataset Information

The project utilizes four datasets:

Providers Data

Contains information about food providers including provider name, type, location, and contact details.

Receivers Data

Contains information about organizations and individuals receiving donated food.

Food Listings Data

Contains details about food items including food type, meal type, quantity, provider information, and expiry date.

Claims Data

Contains information regarding food claims, claim status, receivers, and timestamps.

Dataset Summary:

- Providers Records: 1000
- Receivers Records: 1000
- Food Listings Records: 1000
- Claims Records: 1000

Tools and Technologies Used

Tool	Purpose
MySQL	Database Management & SQL Analysis
Python (Pandas, Matplotlib)	Exploratory Data Analysis
Streamlit	Interactive Dashboard Development
GitHub	Version Control & Project Hosting

SQL Analysis and Business Insights

The project utilized MySQL to perform data analysis and generate actionable insights from the food wastage management dataset. SQL queries involving aggregation functions, JOIN operations, GROUP BY, ORDER BY, and Window Functions were used to analyze food distribution patterns.

Analysis 1: Top Food Providers by Quantity Contributed

```
-- Query 4: Top 10 Providers by Total Food Quantity
-- SELECT
--     p.Provider_ID,
--     p.Name,
--     SUM(f.Quantity) AS Total_Quantity
-- FROM providers p
-- INNER JOIN food_listings f
-- ON p.Provider_ID = f.Provider_ID
-- GROUP BY p.Provider_ID, p.Name
-- ORDER BY Total_Quantity DESC
-- LIMIT 10;
```

Output

Provider_ID	Name	Total_Quantity
709	Barry Group	179
306	Evans, Wright and Mitchell	158
655	Smith Group	150
315	Nelson LLC	142
678	Ruiz-Oneal	140
499	Blankenship-Lewis	124
41	Kelly-Ware	123
161	Campbell LLC	123
262	Bradford-Martinez	121
717	Shepherd and Sons	116

Insight

This analysis identified the providers contributing the highest quantity of food. The results indicate that a small group of providers account for a significant portion of total food donations, making them key contributors to the food distribution network.

Analysis 2: Claim Status Percentage Analysis:

```
-- Query 7: Claim Status Percentage
-- SELECT
--     Status,
--     COUNT(*) AS Total_Claims,
--     ROUND(COUNT(*) * 100.0 /(SELECT COUNT(*) FROM claims),2) AS Percentage
-- FROM claims
-- GROUP BY Status;
```

Output:

Status	Total_Claims	Percentage
Pending	325	32.50
Cancelled	336	33.60
Completed	339	33.90

Insight

The analysis calculated the percentage distribution of claim statuses. This helps evaluate the efficiency of the claim process and identify whether most food listings are successfully claimed or remain pending.

Analysis 3: Provider Ranking Using Window Functions:

```
-- Query 15: Rank Providers by Food Contribution
-- SELECT
--     p.Name,
--     SUM(f.Quantity) AS Total_Quantity,
--     DENSE_RANK() OVER(
--         ORDER BY SUM(f.Quantity) DESC
--     ) AS Provider_Rank
-- FROM providers p
-- JOIN food_listings f
-- ON p.Provider_ID = f.Provider_ID
-- GROUP BY p.Provider_ID, p.Name;
```

Output

Name	Total_Quantity	Provider_Rank
Barry Group	179	1
Evans, Wright and Mitchell	158	2
Smith Group	150	3
Nelson LLC	142	4
Ruiz-Oneal	140	5
Blankenship-Lewis	124	6
Campbell LLC	123	7
Kelly-Ware	123	7
Bradford-Martinez	121	8
Shepherd and Sons	116	9
Hampton-Lee	116	9
Jones, Ortega and Rubio	115	10

Insight

Window functions were used to rank providers based on the quantity of food contributed. This analysis demonstrates advanced SQL concepts and helps identify top-performing providers within the system.

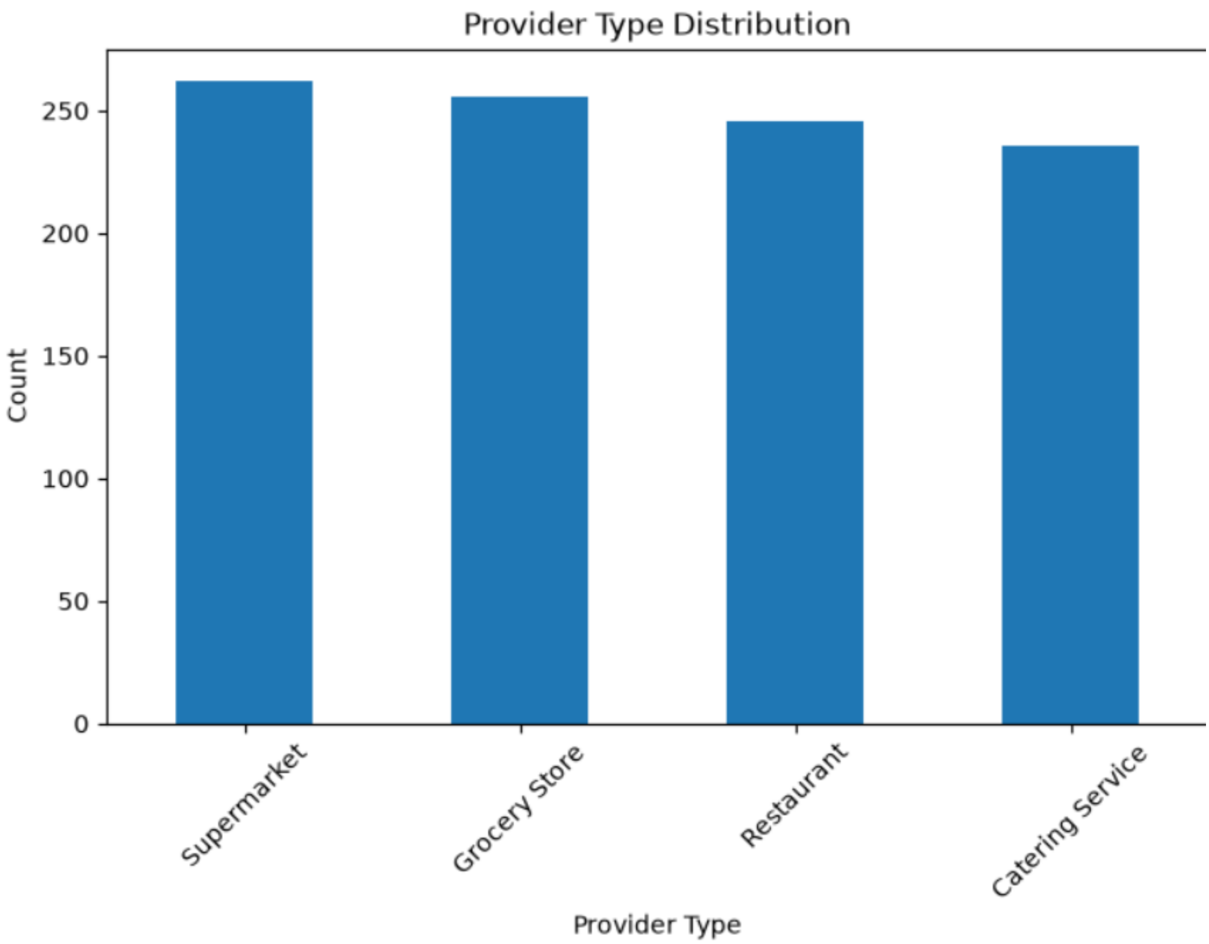
SQL Concepts Applied:

- SELECT Statement
- Aggregate Functions (COUNT, SUM)
- GROUP BY
- ORDER BY
- INNER JOIN
- Subqueries
- Window Functions (DENSE_RANK)
- Data Filtering

Python Exploratory Data Analysis (EDA)

Python libraries such as Pandas and Matplotlib were used to perform Exploratory Data Analysis (EDA) on the food wastage management dataset. The objective was to identify patterns, trends, and distributions within the data to support decision-making and improve food distribution efficiency.

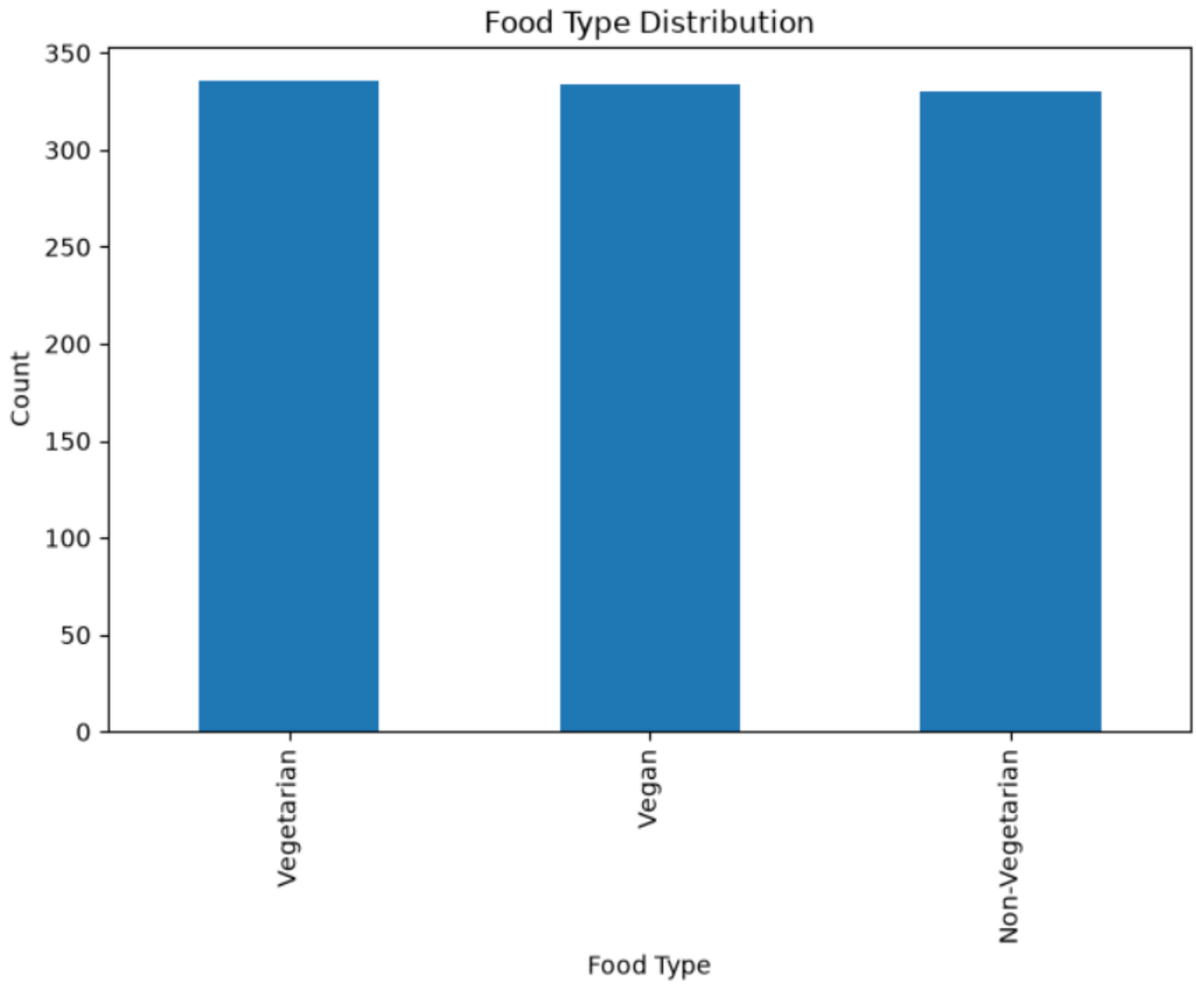
EDA 1: Provider Type Distribution



Insight:

The analysis shows the distribution of food providers across different categories such as supermarkets, grocery stores, restaurants, and catering services. Supermarkets and grocery stores contribute a significant portion of the food listings, highlighting their important role in food donation activities.

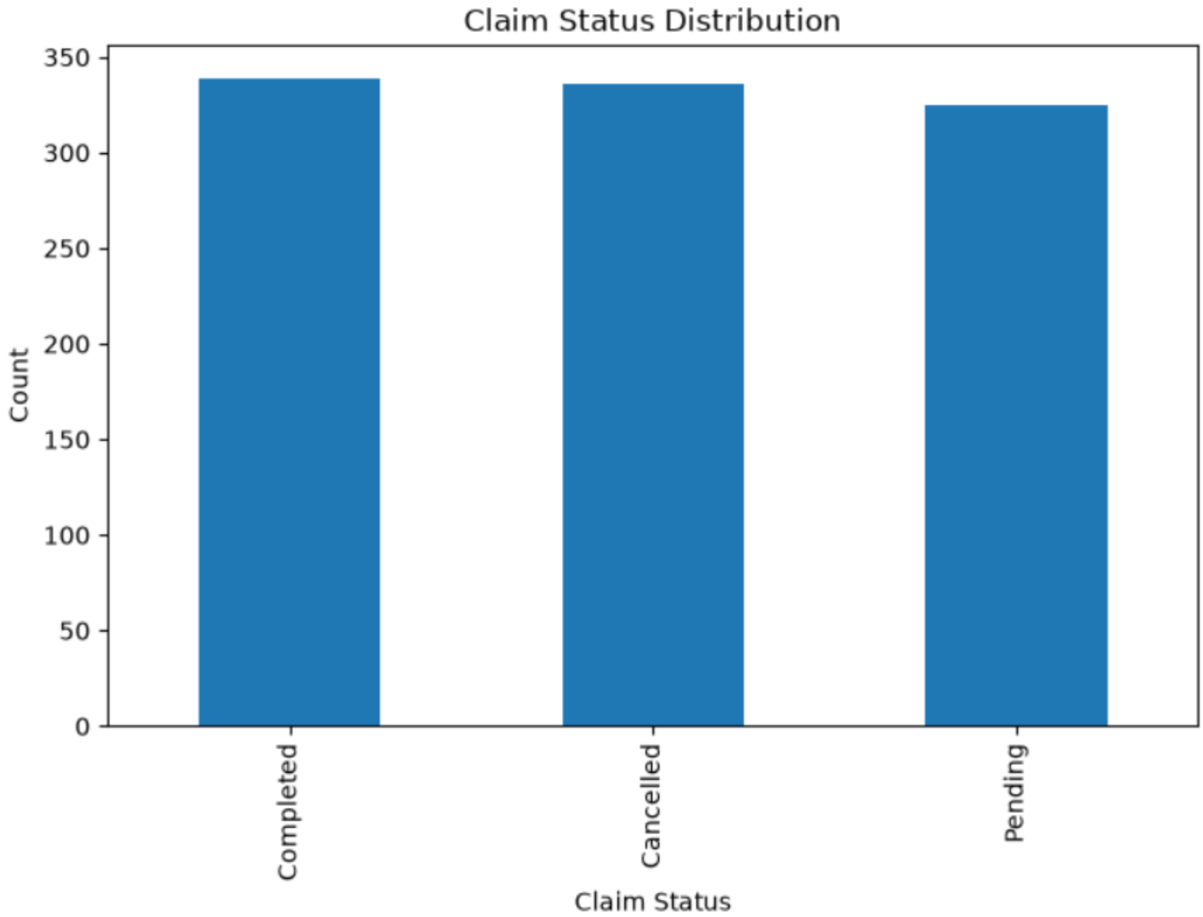
EDA 2: Food Type Distribution



Insight

The food type analysis provides an overview of the different categories of food available in the system. The distribution indicates which food categories are donated most frequently and helps identify donation patterns

EDA 3: Claim Status Distribution



Insight

The claim status visualization illustrates the proportion of completed, pending, and cancelled claims. This helps evaluate the effectiveness of the food distribution process and identify areas for operational improvement.

Key Findings

- Food donations are distributed across multiple provider categories.
- Certain food types are donated more frequently than others.
- Claim status analysis provides insights into distribution efficiency.

Interactive Streamlit Dashboard

An interactive dashboard was developed using Streamlit to provide real-time visualization and analysis of food wastage management data. The dashboard allows users to explore the dataset through dynamic filters and visual charts.

Dashboard Features:

- KPI Cards for Providers, Receivers, Food Listings, and Claims
- Provider Type Filter
- Food Type Filter
- Meal Type Filter
- Claim Status Filter
- Interactive Data Visualizations
- User-Friendly Interface

(A SCREENSHOT OF THE STREAMLIT APP DASHBOARD)

Dashboard Benefits

- Easy monitoring of food donation activities
- Improved visibility into claim and distribution status

- Faster access to business insights
- Interactive exploration of data through filters

Conclusion

The Local Food Wastage Management System successfully demonstrates the application of data analytics techniques to address food wastage challenges. Using Excel, SQL, Python, and Streamlit, meaningful insights were generated from the dataset and presented through an interactive dashboard.

The project highlights how data-driven decision-making can improve food distribution efficiency, increase transparency, and support efforts to reduce food wastage.

Project Links

GitHub Repository

https://github.com/svworks24h-tech/Local-Food-Wastage-Management-System/tree/main

Live Streamlit Dashboard

https://food-wastage-system123.streamlit.app/